IDMA 2024 Problem Set 3

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1a

The formula is

$$((p \to q) \to r) \to ((p \land q) \to r).$$

This formula is a tautology.

To see this, suppose that the right hand side

$$(p \land q) \to r$$

is false. Then we must have

$$p = \text{TRUE}, \qquad q = \text{TRUE}, \qquad r = \text{FALSE}.$$

But then $p \to q$ is true, so

$$(p \to q) \to r$$

is false, since it is $\text{TRUE} \to \text{FALSE}$.

Thus, whenever the consequent of the whole formula is false, the antecedent is also false. Therefore the whole implication cannot be false.

Hence

$$((p \to q) \to r) \to ((p \land q) \to r)$$

is a tautology.

1b

The formula is

$$((p \to q) \land (r \to s)) \leftrightarrow ((p \land r) \to (q \land s)).$$

This formula is neither a tautology nor a contradiction.

First, it is not a tautology. Consider the assignment

$$p = \text{TRUE}, \qquad q = \text{FALSE}, \qquad r = \text{FALSE}, \qquad s = \text{FALSE}.$$

Then

$$p \to q$$

is false, while

$$r \to s$$

is true. Hence the left hand side is false. On the other hand,

$$p \wedge r$$

is false, so

$$(p \wedge r) \to (q \wedge s)$$

is true. Thus the equivalence is false for this assignment.

Second, it is not a contradiction. For example, if

$$p = \text{FALSE}, \qquad q = \text{FALSE}, \qquad r = \text{FALSE}, \qquad s = \text{FALSE},$$

then both $p \to q$ and $r \to s$ are true, so the left hand side is true. Also $p \wedge r$ is false, so the right hand side is true. Therefore the equivalence is true.

Thus the formula is neither a tautology nor a contradiction. A satisfying assignment is

$$p = q = r = s = \text{FALSE}.$$

1c

The formula is

$$((p \wedge \neg r) \vee (q \wedge \neg r)) \to ((p \vee q) \to r).$$

This formula is neither a tautology nor a contradiction.

The left hand side of the implication can be rewritten as

$$(p \wedge \neg r) \vee (q \wedge \neg r) = (p \vee q) \wedge \neg r.$$

The right hand side is

$$(p \vee q) \to r.$$

This is false exactly when

$$p \vee q$$

is true and r is false.

For example, take

$$p = \text{TRUE}, \qquad q = \text{FALSE}, \qquad r = \text{FALSE}.$$

Then the left hand side of the implication is true, but the right hand side is false. Hence the whole formula is false, so the formula is not a tautology.

On the other hand, take

$$p = \text{FALSE}, \qquad q = \text{FALSE}, \qquad r = \text{FALSE}.$$

Then $p \vee q$ is false, so the left hand side of the implication is false. Therefore the whole implication is true.

Thus the formula is neither a tautology nor a contradiction. A satisfying assignment is

$$p = \text{FALSE}, \qquad q = \text{FALSE}, \qquad r = \text{FALSE}.$$

1d

The formula is

$$(p \to (q \vee r)) \vee (\neg(\neg p \vee q) \wedge \neg r).$$

This formula is a tautology.

First,

$$p \to (q \vee r)$$

is equivalent to

$$\neg p \vee q \vee r.$$

Also,

$$\neg(\neg p \vee q)$$

is equivalent to

$$p \wedge \neg q.$$

So the second part of the formula becomes

$$\neg(\neg p \vee q) \wedge \neg r = p \wedge \neg q \wedge \neg r.$$

Therefore the whole formula is equivalent to

$$(\neg p \vee q \vee r) \vee (p \wedge \neg q \wedge \neg r).$$

But

$$p \wedge \neg q \wedge \neg r$$

is exactly the negation of

$$\neg p \vee q \vee r.$$

So the formula has the form

$$A \vee \neg A,$$

where

$$A = \neg p \vee q \vee r.$$

This is always true. Hence the formula is a tautology.

2

The matches are

$$(a) \leftrightarrow (3), \qquad (b) \leftrightarrow (5), \qquad (d) \leftrightarrow (6), \qquad (e) \leftrightarrow (2).$$

Formula (c) does not match any of the listed natural language definitions. Definition (1), the dominating set definition, and definition (4), the connectedness definition, are not matched by any of the predicate logic formulas.

Formula (a)

Formula (a) is

$$\forall u \forall v (E(u, v) \to E(v, u)).$$

This says that whenever there is an edge from u to v, there is also an edge from v to u. Thus every edge appears in both directions.

This is exactly the statement that the graph is undirected. Therefore formula (a) matches definition (3).

$$(a) \leftrightarrow (3).$$

Formula (b)

Formula (b) is

$$\forall u \forall v (E(u, v) \to S(u) \vee S(v)).$$

This says that for every edge (u, v), at least one of the two endpoints is in S. In other words, every edge is incident to at least one vertex in S.

This is exactly the definition of a vertex cover. Therefore formula (b) matches definition (5).

$$(b) \leftrightarrow (5).$$

Formula (c)

Formula (c) is

$$\forall u \forall v \exists w (S(w) \wedge (E(u, w) \vee E(v, w))).$$

This formula does not match any of the listed definitions.

To understand what it says, we can first take $u = v$. Then the formula implies that for every vertex u, there is some vertex $w \in S$ such that

$$E(u, w)$$

holds. So every vertex must have an edge to some vertex in S.

This is close to the idea of a dominating set, but it is not the same. A dominating set only requires every vertex to either be in S, or be a neighbour of a vertex in S. Formula (c) does not allow a vertex to be dominated just by being in S. It still requires an edge from that vertex to some vertex in S.

For example, consider a graph with one vertex x, no edges, and let

$$S = \{x\}.$$

Then S is a dominating set, since the only vertex is in S. But formula (c) is false, because there is no edge $E(x, x)$.

So formula (c) does not match definition (1). It also does not match connectedness, since connectedness is about paths between vertices and does not mention S. Therefore formula (c) has no match among the natural language definitions.

Formula (d)

Formula (d) is

$$\forall u \forall v (E(u, v) \rightarrow ((S(u) \wedge \neg S(v)) \vee (\neg S(u) \wedge S(v)))).$$

This says that whenever there is an edge from u to v, exactly one of u and v is in S. So every edge goes between S and $V \setminus S$, and no edge has both endpoints inside S or both endpoints outside S.

This is exactly the statement that the graph is bipartite with bipartition

$$(S, V \setminus S).$$

Therefore formula (d) matches definition (6).

$$(d) \leftrightarrow (6).$$

Formula (e)

Formula (e) is

$$\forall u \forall v((u \neq v \wedge S(u) \wedge S(v)) \to E(u,v)).$$

This says that if u and v are two distinct vertices in S, then there is an edge from u to v. Thus all vertices in S are neighbours with each other.

This is exactly the definition of a clique. Therefore formula (e) matches definition (2).

$$(e) \leftrightarrow (2).$$

Unmatched definitions

Definition (1) says that S is a dominating set. A formula for this would have to say that every vertex is either in S, or is adjacent to a vertex in S. For example, one possible version would be

$$\forall u(S(u) \vee \exists w(S(w) \wedge E(u,w))).$$

None of the given formulas says exactly this.

Definition (4) says that the graph is connected. This means that for every pair of vertices u and v, there is a path from u to v. None of the given formulas expresses the existence of paths of arbitrary length, so this definition is also unmatched.

3

3a

A republican deck has the ranks

$$A, 2, 3, 4, 5, 6, 7, 8, 9, 10.$$

Thus there are 10 ranks, and each rank has 4 suits. So the deck has

$$10 \cdot 4 = 40$$

cards in total.

A full house consists of three cards of one rank and two cards of another rank.

First choose the rank that appears three times. This can be done in

$$10$$

ways. Then choose the three suits among the four suits for this rank. This can be done in

$$\binom{4}{3}$$

ways.

Next choose the rank that appears twice. Since this rank must be different from the rank used for the three cards, this can be done in

$$9$$

ways. Then choose the two suits among the four suits for this rank. This can be done in

$$\binom{4}{2}$$

ways.

So the number of full houses is

$$10 \cdot \binom{4}{3} \cdot 9 \cdot \binom{4}{2}.$$

The total number of possible 5-card hands from a 40-card deck is

$$\binom{40}{5}.$$

Therefore the probability of getting a full house is

$$\frac{10 \cdot \binom{4}{3} \cdot 9 \cdot \binom{4}{2}}{\binom{40}{5}}.$$

This is equal to

$$\frac{10 \cdot 4 \cdot 9 \cdot 6}{\binom{40}{5}} = \frac{2160}{658008} = \frac{30}{9139}.$$

So the probability is

$$\frac{30}{9139} \approx 0.00328.$$

3b

Jakob's claim is true for full houses.

For a standard deck of cards there are 13 ranks and 4 suits for each rank, so there are 52 cards in total.

The number of full houses in a standard deck is

$$13 \cdot \binom{4}{3} \cdot 12 \cdot \binom{4}{2}.$$

Here we first choose the rank for the three-of-a-kind, then the suits for those three cards, then the rank for the pair, and then the suits for the pair.

The total number of possible 5-card hands from a standard deck is

$$\binom{52}{5}.$$

Thus the probability of getting a full house with a standard deck is

$$\frac{13 \cdot \binom{4}{3} \cdot 12 \cdot \binom{4}{2}}{\binom{52}{5}}.$$

This is

$$\frac{13 \cdot 4 \cdot 12 \cdot 6}{\binom{52}{5}} = \frac{3744}{2598960} \approx 0.00144.$$

For the republican deck, the probability from part 3a was

$$\frac{30}{9139} \approx 0.00328.$$

Since

$$0.00328 > 0.00144,$$

a full house is more likely with the republican deck than with a standard deck.

The intuitive reason is that the republican deck has fewer ranks. This makes it easier for several cards in a 5-card hand to have the same rank. There are fewer cards in total, but the number of suits for each rank is still 4, so repeated ranks become more likely.

4

I use the convention that the rows and columns are ordered as

$$0, 1, 2, 3, 4, 5, 6, 7.$$

For a relation X, the matrix M_X has entry 1 in row i and column j if $(i, j) \in X$, and entry 0 otherwise.

4a

From the directed graph, the relation S contains the pairs

$$\begin{aligned} S = \{&(1,0), (2,0), (3,1), (3,2), (4,0), \\ &(5,4), (5,1), (6,2), (6,4), \\ &(7,3), (7,5), (7,6)\}. \end{aligned}$$

Thus the matrix representation of S is

$$M_S = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 & 1 & 0 \end{pmatrix}.$$

For example, the entry in row 7 and column 3 is 1, since there is an arrow from 7 to 3. The entry in row 0 and column 1 is 0, since there is no arrow from 0 to 1.

4b

The inverse relation I contains all pairs from S, but with the order reversed. Thus

$$(a, b) \in I$$

if and only if

$$(b, a) \in S.$$

So the matrix for I is the transpose of the matrix for S:

$$M_I = M_S^T.$$

Hence

$$M_I = \begin{pmatrix} 0 & 1 & 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

For example, since $(7, 3) \in S$, we get $(3, 7) \in I$. This is why the entry in row 3 and column 7 is 1.

4c

The transitive closure T of I relates two vertices if there is a directed path of length at least 1 from the first vertex to the second vertex in the graph for I.

Using the matrix from part 4b, the reachability sets are

$$
\begin{aligned}
&0 \to \{1,2,3,4,5,6,7\},\\
&1 \to \{3,5,7\},\\
&2 \to \{3,6,7\},\\
&3 \to \{7\},\\
&4 \to \{5,6,7\},\\
&5 \to \{7\},\\
&6 \to \{7\},\\
&7 \to \emptyset.
\end{aligned}
$$

Therefore the matrix representation of T is

$$
M_T = \begin{pmatrix}
0 & 1 & 1 & 1 & 1 & 1 & 1 & 1\\
0 & 0 & 0 & 1 & 0 & 1 & 0 & 1\\
0 & 0 & 0 & 1 & 0 & 0 & 1 & 1\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1\\
0 & 0 & 0 & 0 & 0 & 1 & 1 & 1\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0
\end{pmatrix}.
$$

For instance, row 0 has a 1 in column 7, since in the inverse relation there is a path

$$0 \to 1 \to 3 \to 7.$$

Similarly, row 4 has a 1 in column 7, since

$$4 \to 5 \to 7.$$

4d

The reflexive closure R of T is obtained by adding all pairs

$$(0,0),(1,1),\dots,(7,7)$$

to T. In matrix form, this means changing all diagonal entries of M_T to 1.
So R relates a vertex u to a vertex v if either

$$u = v,$$

or if there is a directed path from u to v in the inverse relation I.
Equivalently, since I is the inverse of S, we have

$$(u,v) \in R$$

if and only if there is a directed path from v down to u in the original graph for S, or $u = v$.

This relation is a reachability relation with paths of length zero or more.

There is also a useful way to interpret the vertices. If the numbers $0, \ldots, 7$ are written in binary, then the arrows in I correspond to adding one new 1-bit. For example,

$$0 = 000, \qquad 1 = 001, \qquad 3 = 011, \qquad 7 = 111.$$

Thus

$$0 \to 1 \to 3 \to 7$$

means that we gradually add bits.

With this interpretation, R is the subset relation on the sets of 1-bits. For example,

$$1 = 001$$

is related to

$$5 = 101,$$

because every 1-bit of 1 is also a 1-bit of 5.

So R behaves like a “less than or equal to” relation on these bit sets. It is reflexive and transitive, and it is a partial order. It is not a total order, since for example 1 and 2 are not comparable.